BTEC Level 3 Applied Science · Unit 1 Physics · Worksheet

Progressive Waves

Original JDScience practice. These questions were written for this resource and are not official exam-board items.

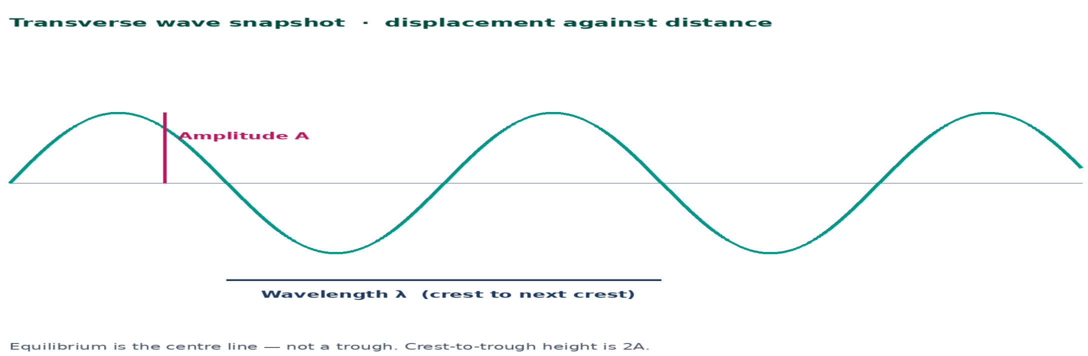
Learning focus: definitions, $v = f\lambda$, standard form and medium change.

A Retrieval

1. Define amplitude, wavelength and frequency.

2. Write the wave equation and give SI units for each quantity, including m s^{-1}.

3. How are period and frequency related?



4. On the snapshot, mark A and λ. How is amplitude read from a displacement–distance graph?

C Calculations

5. $f = 4.0$ Hz, $\lambda = 0.80$ m. Calculate v.

Working

6. A radio wave has $f = 200$ MHz. Take $c = 3.00 \times 10^{8}$ m s^{-1}. Calculate λ.

Working

7. A wave travels 48 m in 0.16 s. If $\lambda = 2.0$ m, calculate f.

Working

D Exam-style practice

8. Define amplitude and wavelength. [2]

9. A note of 510 Hz travels at 340 m s^{-1}. Calculate λ. [2]

Working

10. Explain why air particles do not travel from a loudspeaker to a listener. [3]

E Challenge

11. A displacement–time trace has 5 complete cycles in 20 ms. The matching snapshot shows 4.0 cm between adjacent crests. Calculate f, T, λ and v. Show unit conversions.

Working

12. Which quantity stays the same when a wave enters a new medium? Explain what happens to v and λ.